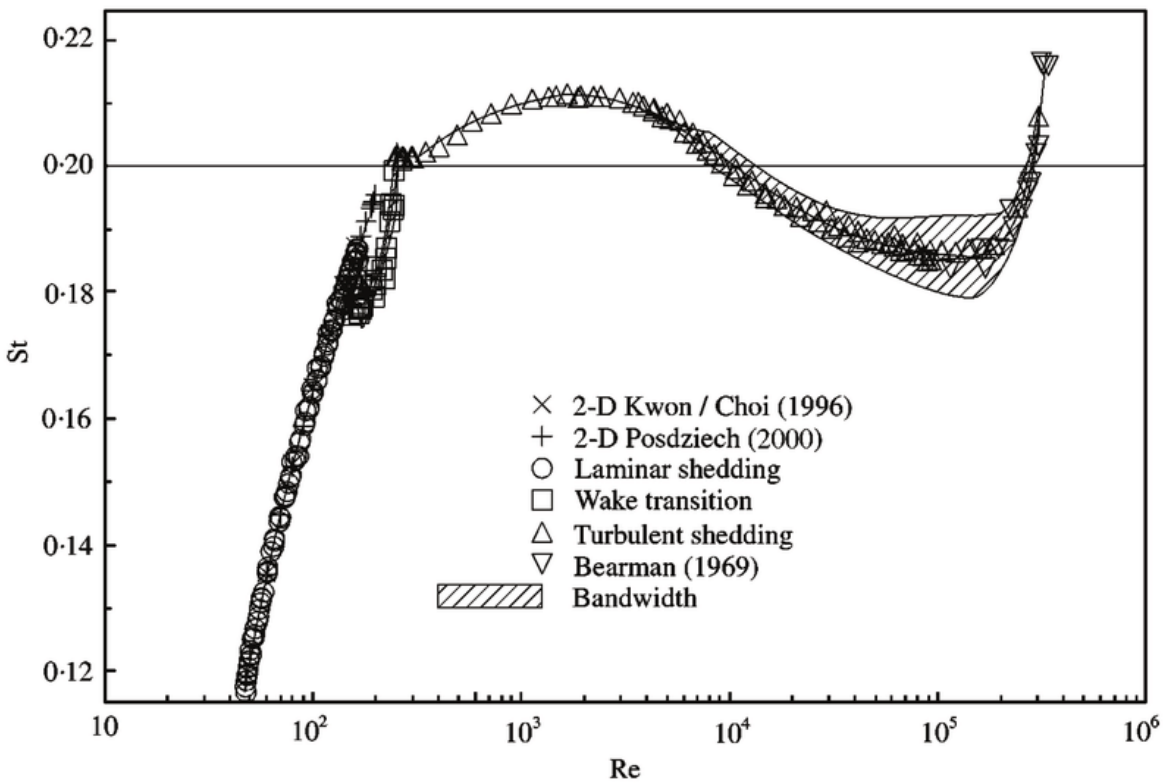
AE 462
Lab 5
Lab Session: Wednesday, March 11, 2026
Due: Wednesday, March 25, 2026, 9am CST

This laboratory assignment will focus on using CFD simulations of unsteady flow fields. The items required for your assignment submission are given here, alongside helpful instructions for completing this lab. As with previous assignments, all plots and presentations of data should include units, axis titles, and legends where appropriate. Provide your responses in a single PDF file. Upload any scripts, spreadsheets, or other work used during data analysis. Also upload the .geo file you used to generate your mesh in Gmsh.

Report Instructions:
Using CFD-generated data from this exercise, provide a response to the following:

1) **Transient Time**
    a. In a separate plot for each Reynolds number case, provide the unsteady C_l signals for the cylinder as a function of time (Cur_Time variable), from $t = 0$ to $t = 5$ seconds. In these plots, provide a vertical line that marks the estimated end of the transient time of the flow.
    b. Comment on the changes you observe with increasing Reynolds number in:
        i. The amplitude of the unsteady variation in C_l.
        ii. The changes in transient time.
2) **Strouhal Number**
    a. Calculate the power spectral density of the fluctuations in C_l. For each Reynolds number, provide a plot showing the resulting power spectral density as a function of frequency. Use a $\log_{10}$ basis to plot the power spectral density on the y axis (unless plotting values in dB/Hz). There are several standard methods for calculating power spectral density profiles using Python or Matlab. Note that $dt = 0.02$ is the time step assumed, which also corresponds to a sample rate of $f_s = 50$ Hz.
    b. Identify the frequency of peak power in the spectrum for each Reynolds number. Determine the corresponding Strouhal number for Re = 100, 200, and 400.
    c. A plot, showing a collection of Strouhal numbers for vortex shedding from a circular cylinder is given below. Comment on any consistencies or inconsistencies between your results and those found from prior research.
3) **Vortex Spacing**
    a. Provide the three snapshots of the velocity magnitude field acquired across the three Reynolds number conditions.
    b. From the plots provided, estimate the vortex spacing, in meters, produced for each Reynolds number case. Here, assume that the vortex spacing represents the streamwise distance between vortices shed from one side of the cylinder (e.g., physical distance from vortex (1) shed from the upper side of the cylinder, and vortex (2) also shed from this location).


0·22
0·20
0·18
0·16
0·14
0·12
St
10
10^2
10^3
10^4
10^5
10^6
Re
2-D Kwon / Choi (1996)
2-D Posdziech (2000)
Laminar shedding
Wake transition
Turbulent shedding
Bearman (1969)
Bandwidth

Lab Instructions:

Organizing files

Create a new folder within your U:\AE 462 directory called "Lab 5". Download the "unsteady_cylinder.cfg" and "Cylinder_Omesh_Template.geo" files. Place these files in your U:\AE 462\Lab 5 directory.

Creating a Mesh

The .geo template contains a notional outline of the flow domain for a circular cylinder in a freestream. Using this outline, create a structured mesh for this flow. Begin by creating two plane surfaces that enclose the upper and lower halves of the field mesh. Placement the mesh nodes through the transfinite curve definition routine. Recommended values include:

> # nodes in tangential direction: 200, evenly spaced
> # nodes in wall-normal direction: 360, with progression biased towards cylinder surface
> Progression = 1.03 provides a good balance of near-field and far-field resolution

Create your structured mesh, export the mesh in .su2 format, and be sure to leave checked "Save all elements". The unsteady_cylinder.cfg file is configured to read a mesh file named "Cylinder.su2", though if you use a different name make sure this file name is consistent with "MESH_FILENAME= ###" in your unsteady_cylinder.cfg file before running (### is the mesh file name).

Running Re = 100

The default case implemented in the unsteady_cylinder.cfg file includes a freestream Reynolds number of 100. This vicinity of Reynolds numbers is the laminar vortex shedding region and, for this reason, we are using an average form of the governing equations that assume laminar flow and omitting the use of a turbulence model.

```
SOLVER= NAVIER_STOKES
KIND_TURB_MODEL= NONE
```

The default unsteady_cylinder.cfg file also uses a 2nd order finite differencing strategy to couple the physical and pseudo time derivatives of the dual time stepping routine. The unsteady_cylinder.cfg file is also configured to assume convergence of the outer (physical) time solution when the windowed residual decreases below 10^{-5}.

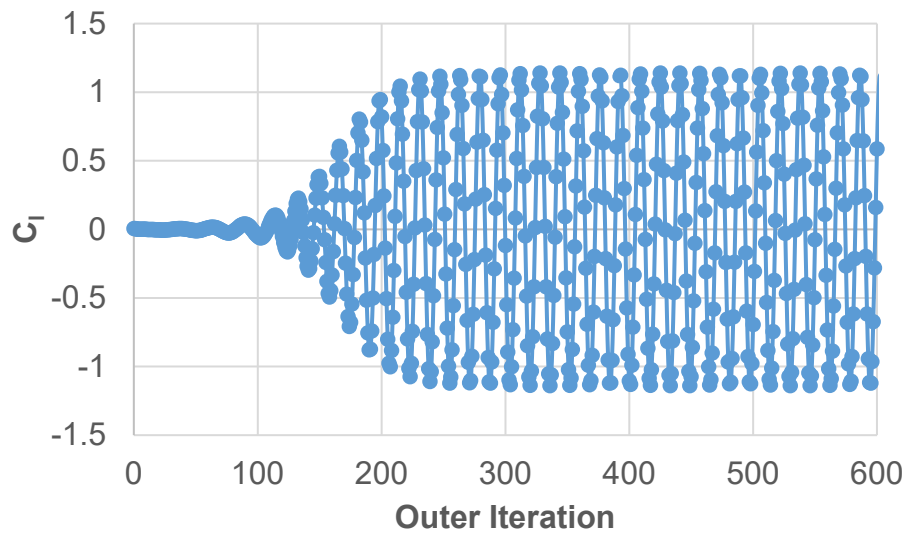
Run SU2 in parallel using your O-mesh and baseline .cfg file.

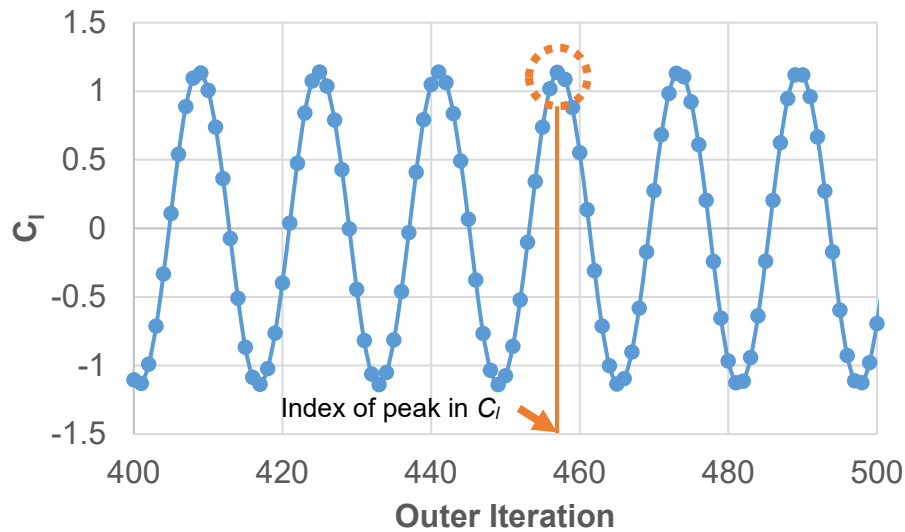
```
mpiexec -n 8 "%SU2_RUN%\SU2_CFD" unsteady_cylinder.cfg
```

or similar (above example uses 8 processors). One the solution converges, save the "history.csv" file as "history_100.csv" for later analysis.

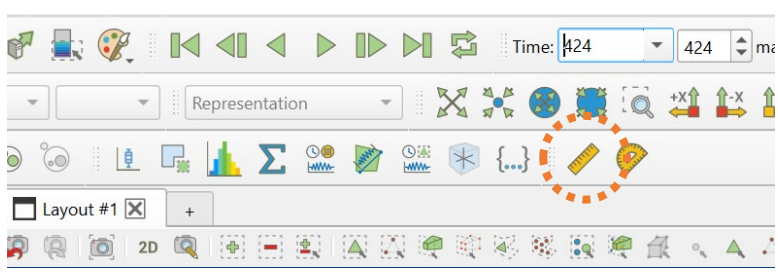
Additionally, when plotting the history of C_l vs. Outer Loop step, you will see a clear maximum value produced across the oscillation cycle. Identify the time step associated with one of these

maxima, somewhere in the middle of the total record length of the simulation (e.g., after 300+ outer iterations).





For this time step, plot the velocity magnitude field in Paraview. Add a ruler to the flow by clicking on the ruler icon. Set the start and end points to be (0, 0, 0) and (10, 0, 0), respectively. Zoom and pan your view to see the cylinder and wake domain across the length of the ruler. Take a snapshot of your flow field for this Reynolds number and save for later comparisons.



Running Other Reynolds Numbers
Repeat this process for Reynolds numbers of 200 and 400. After converging to a stable limit-cycle oscillation, save the history.csv files and capture a snapshot of the flow in Paraview corresponding to an iteration number where a maximum in C_l is produced on the cylinder. Place a ruler, with the same start and end points used previously, and capture a snapshot for these additional Reynolds number cases.

Calculating Strouhal Number
Using your history.csv files, generate a power spectral density for the C_l signal. From this power spectrum, determine the primary frequency of the unsteady vortex shedding process. This frequency can be expressed in a dimensionless form, known as the Strouhal number (St). With this frequency, determine the value of St produced for Re = 100, 200, and 400. You can calculate St using the following:

$$St = \frac{fD}{U}$$

where f is the frequency determined from the power spectrum, D is the diameter of the cylinder (2 m), and U is the freestream velocity (34.03 m/s).

You now have the CFD-generated data needed to respond to the items required for the submission.